

scripts/run-challenge-matrix.sh

— User Guide

Last verified: 2026-08-26 (LVA-161 — --test-timeout pass-through; the CLI's 10m default was killing whole-module sweeps and destroying every class's results) **Inheritance:** HelixConstitution §11.4.18 + Lava §6.AE.6 + §6.X (Container-Submodule Emulator Wiring) + §6.I (Multi-Emulator Container Matrix)

Overview

Operator entry point for §6.AE gate-mode Challenge Test runs. Pre-bakes the §6.AE.2 minimum AVD matrix (API 28 / 30 / 34 / latest stable × phone + tablet) and delegates to submodules/containers/cmd/emulator-matrix --runner=containerized per §6.X.

LVA-014 durable device-gate fixes (2026-07-26)

The 2026-07-04 "device gate boot hang" was diagnosed as an **AVD-name mismatch**: the baked emulator images (ghcr.io/vasic-digital/lava-android-emulator:apiNN-x86_64) bake exactly ONE AVD named default, but the runner passed the §6.AE.2 matrix names (CZ_API34_Phone, ...) verbatim. The image entrypoint exited in ~4s ("AVD not found"), the container's --rm reaped the log, and WaitForBoot polled the dead forwarded port to its deadline, misreporting the 4s exit as a boot timeout. Three durable fixes:

1. **AVD-name resolution** (submodules/containers/pkg/emulator/avdresolve.go, wired into Containerized.Boot) — the runner queries the AVDs actually baked into each image (avdmanager list avd inside a throwaway --rm --entrypoint avdmanager container) and maps the requested name:api:form entry to a real baked AVD. Exact name match → used verbatim (the proven --avds "default:34:phone" workaround is byte-compatible). Api-level match → the baked name is substituted with a stderr note (the

requested name was advisory). No match → Boot fails FAST, before any container is launched, naming the available baked AVDs.

2. **Container liveness in WaitForBoot** (submodules/containers/pkg/emulator/containerized.go) — every poll iteration first checks `podman/docker inspect` that the emulator container is still running; on exit it fails immediately with the container's captured logs (`<runtime> logs --tail 100`, fetched before `--rm` reaps them; an already-reaped container is reported honestly). EMU-1 semantics are preserved: a Containerized without a Boot-set container name performs no liveness exec, and every liveness exec is bound to the WaitForBoot deadline context.
3. **Emulator-image preflight** (this script, containerized runner only) — before building the APK, the script instantiates the `{api}` image template per distinct api level in the matrix, checks each image with `<runtime> image inspect`, and pulls the missing ones with live progress. A pull failure is an honest operator-facing error (exit 1) that names the image, shows the pull error tail, and prints the local-build fallback command from `submodules/containers/pkg/emulator/Containerfile` — never a silent skip.

The default `--container-image` is now the template `ghcr.io/vasic-digital/lava-android-emulator:api{api}-x86_64`; the runner substitutes `{api}` with each AVD's api level at Boot time, so one reference covers the whole multi-api matrix. A `--container-image` without `{api}` is used verbatim for every AVD (the pre-LVA-014 single-image behavior).

Honest pre-flight

The script DETECTS the §6.X-debt darwin/arm64 host gap (no `/dev/kvm` available to podman containers; macOS HVF not exposed to podman) and, when detected, REFUSES to claim a successful gate run. Instead it:

1. Validates arguments + matrix minimum
2. Writes `<evidence-dir>/host-preflight.json` with the host-gap classification
3. Exits 2 (NOT 0) — gate-host ineligible

Per §6.J/§6.L: no false-pass; honest unblock report. Real gate runs require a Linux x86_64 + KVM host.

Usage

Run ALL Challenges on the \$6.AE.2 minimum matrix

```
bash scripts/run-challenge-matrix.sh
```

Run a specific Challenge

```
bash scripts/run-challenge-matrix.sh \  
  --test-class  
    lava.app.challenges.Challenge01AppLaunchAndTrackerSelectionTest
```

Add TV-class AVD (when feature touches TvActivity)

```
bash scripts/run-challenge-matrix.sh --add-tv
```

Add foldable AVD

```
bash scripts/run-challenge-matrix.sh --add-foldable
```

Override "latest stable" API level

```
bash scripts/run-challenge-matrix.sh --latest-api 36
```

Skip APK rebuild

```
bash scripts/run-challenge-matrix.sh --no-build
```

*# Override the matrix entirely with explicit AVDs (target EXISTING
 host AVDs).*

Use on a macOS host-direct+HVF run where the default CZ_API
 images are not provisioned.*

```
bash scripts/run-challenge-matrix.sh --avds  
  "CZ_API35_Phone_Fresh:35:phone" --no-build
```

*# Single-AVD smoke run against the locally present api34 image
 (the requested*

*# name is advisory – the runner resolves the image's baked AVD,
 e.g. "default")*

```
bash scripts/run-challenge-matrix.sh --avds  
  "CZ_API34_Phone:34:phone" --no-build \  
  --test-class lava.app.challenges.Challenge00ColdStartTest
```

*# Pin one explicit image for every AVD (pre-LVA-014 single-image
 behavior)*

```
bash scripts/run-challenge-matrix.sh \  
  --container-image ghcr.io/vasic-digital/lava-android-  
    emulator:api34-x86_64
```

ALSO invoke the 11 HelixQA Challenge scripts (Option 1 wiring)

```
bash scripts/run-challenge-matrix.sh --include-helixqa
```

Inputs

Arg	Description
--test-class <fqcn>	

Arg	Description
	Specific Challenge to run (default: all under <code>lava.app.challenges</code>)
<code>--evidence-dir <dir></code>	Output directory (default: dated under <code>.lava-ci-evidence/</code>)
<code>--no-build</code>	Skip the <code>:app:assembleDebug</code> step
<code>--avds</code> <code>"<name:api:form[,...]>"</code>	Replace the §6.AE.2 default matrix entirely with an explicit comma-separated AVD list. Use to target existing host AVDs (e.g. <code>CZ_API35_Phone_Fresh:35:phone</code> on a macOS host-direct+HVF run where the default <code>CZ_API*</code> images are not provisioned). When set, <code>--latest-api</code> / <code>--add-tv</code> / <code>--add-foldable</code> are ignored. A sub-minimum list is a development-iteration run, not a §6.AE.2-conformant gate matrix. NOTE: the Containers CLI still resolves each AVD's system image through <code>tools/lava-containers/vm-images.json</code> (<code>--image-manifest</code>); the manifest must contain a matching image id (e.g. <code>android-35-phone</code>) or the runner fails with <code>no image with id</code> .
<code>--latest-api <N></code>	Override the "latest stable" API level (default: 36; ignored when <code>--avds</code> is set)
<code>--container-image <ref></code>	Containerized-runner image. Default: <code>ghcr.io/vasic-digital/lava-android-emulator:api{api}-x86_64</code> — the <code>{api}</code> token is substituted per AVD api level by the runner (LVA-014 fix #1). A ref without <code>{api}</code> is used verbatim for every AVD.
<code>--container-runtime <podman docker></code>	Container runtime CLI (default: <code>podman</code>).
<code>--extra-apk <path></code>	Additional APK installed on each AVD AFTER the client APK, before instrumentation runs. Repeatable — pass more than once to install several companion APKs. Forwarded verbatim, in order, to the Containers CLI's own <code>repeatable --extra-apk</code> flag (<code>MatrixConfig.ExtraAPKPaths</code>). Use for

Arg	Description
	Challenges needing a second app on the SAME device (e.g. Challenge72's on-device api-app companion for same-device mDNS discovery). Added for the §6.X gate wiring that closes the two-app-on-one-emulator capability gap.
--add-tv	Add a TV-class AVD to the matrix
--add-foldable	Add a foldable AVD
--include-helixqa	ALSO invoke scripts/run-helixqa-challenges.sh (the 11 HelixQA Challenge scripts per Option 1 wiring). OFF by default so existing matrix runs are unaffected. HelixQA runs on the HOST BEFORE the AVD matrix → independent of §6.X-debt darwin/arm64 gate-host gap. HelixQA wrapper evidence lands at <evidence-dir>/helixqa/. Non-zero HelixQA exit promotes the final aggregate exit code (matrix exit dominates if both fail). See docs/scripts/run-helixqa-challenges.sh.md for full details.

LVA-161 — --test-timeout pass-through (2026-08-26)

The defect

submodules/containers/cmd/emulator-matrix defaults --test-timeout to **10 minutes** (cmd/emulator-matrix/main.go:123). That budget covers the **TEST step only** — cold boot is timed separately and reported as boot_seconds — and this script never set it, so every run inherited the default.

A whole-module Challenge sweep is a **single gradle invocation** covering every selected class, so 10 minutes is far below the real workload. Measured: the 2026-08-26T14-09-17Z run was killed at test_seconds=600.02 with signal: killed, having completed **81 of 104 tests**.

What makes this worse than a truncated run: gradle writes its JUnit XML at the **end** of the invocation. A kill therefore destroys the results of **every** class in the sweep, including the ones that had already passed — so the matrix reports the whole selection as failed when nothing about the product was wrong. That is a verdict determined by a clock, not by the code under test.

What changed here

This script gained a `--test-timeout <duration>` argument that is forwarded verbatim to `emulator-matrix --test-timeout` when set, and omitted entirely when empty (preserving the CLI default). That is the whole change: this script does **not** compute a value.

Callers MUST size it to the selected workload. The pipeline's Challenge phase (`scripts/pipeline/phase-02-test-challenge.sh`) derives it as $300s + 45s \times \text{<selected class count>}$ — a fixed overhead plus a per-class budget — and prints the derivation with the class count before invoking. Both terms are overridable (`LAVA_PIPELINE_CHALLENGE_TEST_TIMEOUT_OVERHEAD_S`, `LAVA_PIPELINE_CHALLENGE_TEST_TIMEOUT_PER_CLASS_S`), and an explicit `LAVA_PIPELINE_CHALLENGE_TEST_TIMEOUT` wins verbatim. Operators invoking this script directly for a large selection should pass `--test-timeout` themselves rather than accept 10m.

Reading a timeout kill honestly

A run killed at the timeout is **not** a test failure and must not be recorded as one. The distinguishing evidence is `test_seconds` sitting within a rounding error of the configured budget together with `signal: killed`, and a JUnit XML that is absent rather than reporting failures. The remedy is to raise `--test-timeout` (or reduce the class selection) and re-run to obtain a real verdict — never to re-interpret the kill as a product defect.

Inputs (addition)

Arg	Description
<code>--test-timeout <duration></code>	Forwarded to <code>emulator-matrix --test-timeout</code> (Go duration syntax, e.g. 2400s, 40m). Empty = the CLI default of 10m, which is too small for a whole-

Arg	Description
	module sweep. Applies to the TEST step only; cold boot is budgeted separately by <code>--boot-timeout</code> .

§6.AE.2 mandatory minimum matrix

CZ_API28_Phone:28:phone
 CZ_API30_Phone:30:phone
 CZ_API34_Phone:34:phone
 CZ_API34_Tablet:34:tablet
 CZ_API<latest>_Phone:<latest>:phone

Plus TV / foldable when explicitly requested by `--add-tv` / `--add-foldable`.

Sub-minimums are permitted for development iteration; the gate row's gating: `true` flag is only set when the full minimum is satisfied + every config dimension (theme/locale/density per §6.AE.2) is covered.

Outputs

- `<evidence-dir>/host-preflight.json` — host-gap classification (always written)
- `<evidence-dir>/real-device-verification.{md,json}` — per-AVD attestation rows (only on Linux x86_64 + KVM gate-host)
- `<evidence-dir>/helixqa/helixqa-attestation.json` + per-script logs — only when `--include-helixqa` is set

Exit codes

Code	Meaning
0	Matrix completed; all AVDs passed
1	At least one AVD failed boot OR at least one test failed (real Containers CLI exit), OR the LVA-014 image preflight failed (missing image + pull failure — honest error with local-build fallback)
2	Gate-host ineligible (darwin/arm64 OR no KVM); honest unblock-needed report written

Cross-references

- `submodules/containers/cmd/emulator-matrix` (the underlying runner)
- `tools/lava-containers/vm-images.json` (matrix manifest)
- `scripts/run-emulator-tests.sh` (older sister-glue; same delegation, different default arguments)
- `scripts/run-helixqa-challenges.sh` + `docs/scripts/run-helixqa-challenges.sh.md` (the HelixQA wrapper invoked by `--include-helixqa`)
- `docs/plans/2026-05-16-helixqa-integration-design.md` (Option 1 wiring design)
- `.lava-ci-evidence/sixth-law-incidents/2026-05-13-emulator-container-darwin-arm64-gap.json` (the standing §6.X-debt incident)
- `Lava CLAUDE.md` §6.AE + §6.X + §6.I